Shuvo *et al. Agricultural Products Processing and Storage* (2026) 2:26
https://doi.org/10.1007/s44462-026-00061-9

RESEARCH

Open Access

Image-based classification of Bangladeshi rice varieties using deep convolutional neural networks

Syme Al-Azad Shuvo[1], Khalilur Rahman[1], Abdullah Al Naseeh Chowdhury[1], Gourab Roy[1], Nafiz Nahid[1], Mahmud Hasan[2], Md. Abu Naser Mojumder[1*], Md. Janibul Alam Soeb[2] and Md. Fahad Jubayer[3*]

Abstract

This research investigates the application of deep convolutional neural networks (CNNs) for classifying 5 prominent rice varieties in Bangladesh: Basmati, Chinigura, Jirashail, Kataribhog, and Paijam. Rice is a significant agricultural product in Bangladesh. Proper categorization of rice varieties is vital for quality management, preventing adulteration, and preserving customer confidence. This research leverages advanced CNN architectures (VGG16, VGG19, ResNet50, Xception, DenseNet) and a custom model, BDriceNetworkV0, to automate the classification process. A balanced dataset comprising 15 000 images of rice grains was collected under controlled conditions to train the models. Performance metrics, including accuracy, precision, recall, and F1-score, were used to evaluate the models. VGG19 was identified as the superior performer (accuracy 99.73%). The custom model, BDriceNetworkV0, exhibited potential for resource-constrained applications (accuracy 98.60%); however, with slightly reduced precision. The findings highlight the feasibility of applying deep learning for rice classification, with potential benefits for enhancing quality, reducing operational costs, and improving food security in Bangladesh. This study offers valuable insights into the application of AI in agriculture, particularly in grain classification.

Keywords Computer vision, Rice variety, Food processing, Deep learning, Grain classification

Introduction

In recent years, the combination of image processing and machine learning (ML) has instigated a significant transformation in agriculture, enhancing precision, efficiency, and automation in a traditionally labor-intensive sector. From detecting and identifying plant diseases to evaluating quality, image-based computational approaches are reshaping the landscape of agricultural technology [1]. Among these developments, deep learning (DL) is distinguished by its exceptional ability to perform intricate visual tasks, such as object detection, localization, and classification, with near-human precision. As global agricultural systems contend with increasing demand and dwindling resources, the incorporation of artificial intelligence (AI) has become a crucial facilitator of smart and precision farming [2].

Rice, essential for more than half the world's population, plays a vital role in agriculture, culture, and the economy, especially in Asia. With a history of thousands of years, rice cultivation has over 100 000 varieties suited to different zones [3]. In Bangladesh, rice is key to food security and rural livelihoods, grown across three seasons on over 11.7 million hectares, about 77% of cropped land

*Correspondence:
Md. Abu Naser Mojumder
abu.naser84@gmail.com
Md. Fahad Jubayer
fahadbau21@hotmail.com; jubayer.fet@sau.ac.bd
[1] Department of Computer Science and Engineering, Sylhet Engineering College, Sylhet 3100, Bangladesh
[2] Department of Farm Power and Machinery, Sylhet Agricultural University, Sylhet 3100, Bangladesh
[3] Department of Food Engineering and Technology, Sylhet Agricultural University, Sylhet 3100, Bangladesh

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

[4]. However, identifying rice varieties quickly, accurately, and at scale is challenging amid over 40 000 global types classified by grain size [5]. Correct identification matters for supply, markets, consumer trust, and preventing fraud [6]. Traditional methods, mainly manual, are unreliable, subjective, time-consuming, and error-prone [2]. As a result, the demand for automated, intelligent solutions has never been more urgent. The progress of computer vision and AI, especially DL, has created new frontiers in agricultural automation and grain classification.

Deep convolutional neural networks (CNNs), fundamental to contemporary computer vision, provide an elegant solution to this issue. Through the acquisition of hierarchical information from images, CNNs may distinguish subtle nuances in grain morphology, texture, and chromaticity, characteristics that are frequently imperceptible to the unaided human eye [2, 7, 8]. With the advent of transfer learning and robust pre-trained architectures such as ResNet, Visual Geometry Group (VGG), Xception, AlexNet, and DenseNet, performance benchmarks in agricultural image classification have reached new heights [9]. Studies have repeatedly shown the advantages of these models in real-world applications. For instance, Kelishami et al. [10] utilized ResNet with fine-tuning to accurately classify Persian rice varieties using images captured on mobile phones. The study leveraged CNNs, specifically fine-tuning a ResNet model, to accurately classify rice varieties from images captured with consumer-grade cell phones. The research not only showcased the technical feasibility of mobile-based rice variety detection but also emphasized the broader implications for consumer experiences in food selection and agricultural technology. Similarly, Sharma et al. [11] demonstrated that DL models consistently outperformed conventional ML approaches across various metrics. ResNet50 performed well in this study compared to other DL models, including CNN, Deep CNN, AlexNet2, Xception, Inception V3, and DenseNet121. Razavi et al. [12] highlighted how transfer learning with ResNet can successfully differentiate between visually similar rice cultivars, while Priya et al. [13] introduced DeepRice-Transfer. This model underscores the robustness of transfer learning in crop classification. Beyond classification, CNNs have also demonstrated promise in tasks such as rice phenotyping and disease detection [14], further underscoring their versatility in agricultural domains. While general rice classification has been explored, a comparative benchmark of modern CNN architectures specifically for the commercially vital Bangladeshi varieties–Basmati, Chinigura, Jirashail, Kataribhog, and Paijam–remains absent from the literature.

Therefore, the current study aims to examine the effectiveness of DL architectures in classifying 5 popular Bangladeshi rice varieties: Basmati, Chinigura, Jirashail, Kataribhog, and Paijam. These varieties are not only agriculturally significant but also have particular culinary and commercial value. Accurate identification of these grains is crucial for maintaining quality standards and combating counterfeiting in local and international markets. In this context, our research pursues 2 core objectives: 1) to evaluate the classification performance of widely adopted CNN models (VGG16, VGG19, ResNet50, Xception, and DenseNet, under controlled imaging conditions using a moderately sized dataset, and 2) to compare these pre-trained models against a custom-built DL framework, BDriceNetworkV0) to assess their relative strengths for real-world deployment. The significance of this work lies not only in its methodological contribution but also in its practical applications. Using such models in industrial settings could transform grain sorting, cut operational costs, and maintain consistent quality control. Moreover, these approaches are inherently scalable and can be adapted for edge computing, enabling real-time analysis and broader applicability across grain types and feature modalities (e.g., aroma, nutritional content). As Chen et al. [15] point out, deep transfer learning is highly generalizable and holds immense potential for cross-domain applications in agriculture and beyond. In conclusion, this study contributes to the expanding frontier of agricultural AI by rigorously benchmarking CNN-based architectures for a real-world problem of national importance. Through a balanced evaluation of both established and custom models, it sheds light on the possibilities of AI-driven grain classification and its broader role in supporting food security, supply chain transparency, and technological innovation in Bangladesh's agriculture sector.

Methodology

Dataset details

To support robust research in rice grain classification, we curated a high-quality image dataset comprising 5 widely cultivated Bangladeshi rice varieties: Basmati, Chinigura, Jirashail, Kataribhog, and Paijam. The rice samples were sourced from 4 different regions of Bangladesh, including Sylhet, Mymensingh, Dinajpur, and Gopalganj, ensuring geographic diversity and varietal authenticity. While slight variations in grain size were observed across regions, the distinct morphological shape unique to each variety remained consistent. Image acquisition was conducted under controlled lighting conditions using a Canon EOS M50 Mark II DSLR camera equipped with an 18–45 mm kit lens (24.1 MP APS-C CMOS sensor). Each grain image was captured individually on a custom-built imaging stage to standardize viewpoint, illumination, and focus. An initial set of 15 852 images was collected and

manually filtered to ensure high quality. Exclusion criteria included: 1) grains with visible cracks or structural damage, 2) blurred images caused by focus errors, and 3) images with uneven lighting or shadows. This ensured the dataset represented ideal morphological features. This process resulted in a clean and balanced dataset of 15 000 high-resolution images, with 3 000 samples allocated to each rice variety. To facilitate practical model training and evaluation, the dataset was systematically partitioned into three subsets: training, validation, and testing (Table 1).

Table 1 Size of the dataset for each classification approach of the rice variety

Class	Train	Validation	Test	Total
Basmati	2 400	300	300	3 000
Chinigura	2 400	300	300	3 000
Jirashail	2 400	300	300	3 000
Kataribhog	2 400	300	300	3 000
Paijam	2 400	300	300	3 000

Each class contributes 2 400 images for training, 300 for validation, and 300 for testing, ensuring both uniform class distribution and unbiased performance assessment. The representative samples of each rice class are illustrated in Fig. 1. This carefully curated and balanced dataset provides a solid foundation for training DL models, enabling both intra-class feature extraction and inter-class discrimination. It also serves as a benchmark for future research in automated rice classification and grain quality assessment.

Training and testing procedure

The study framework (Fig. 2) involves a systematic pipeline designed to ensure accurate and reliable classification of 5 Bangladeshi rice grain varieties. The process begins with acquiring a curated image dataset that ensures diversity and comprehensiveness across

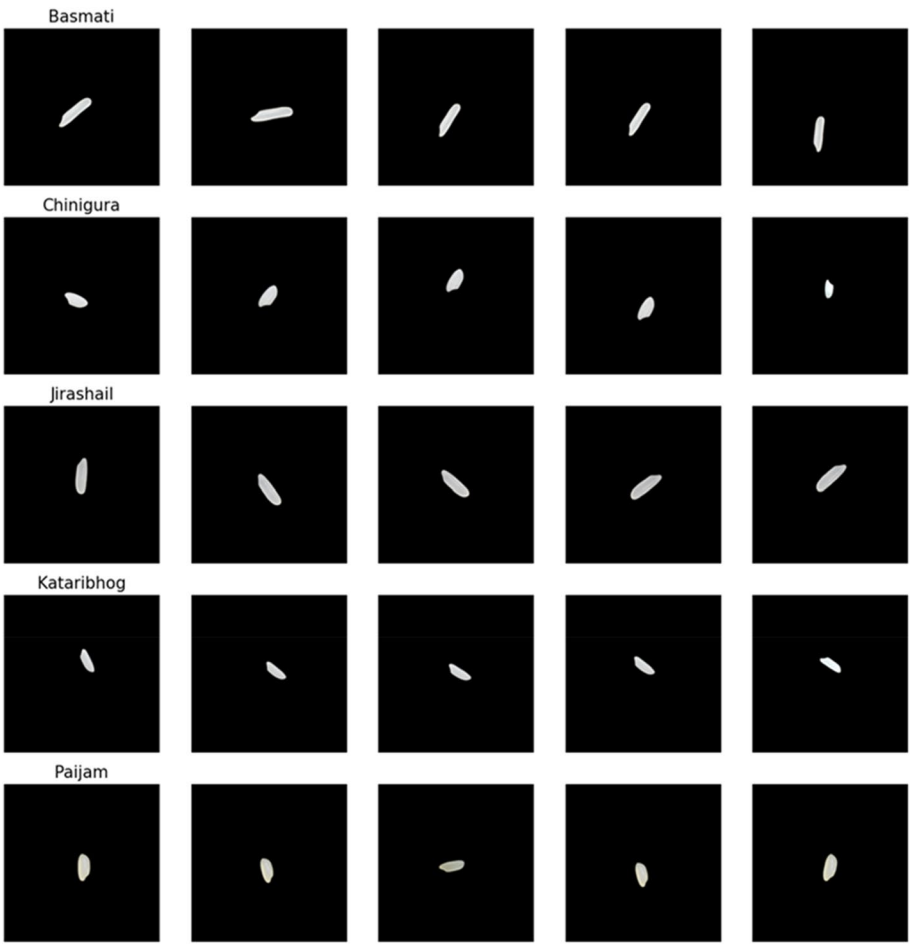


Fig. 1 Sample pictures from each class of rice variety used in the study

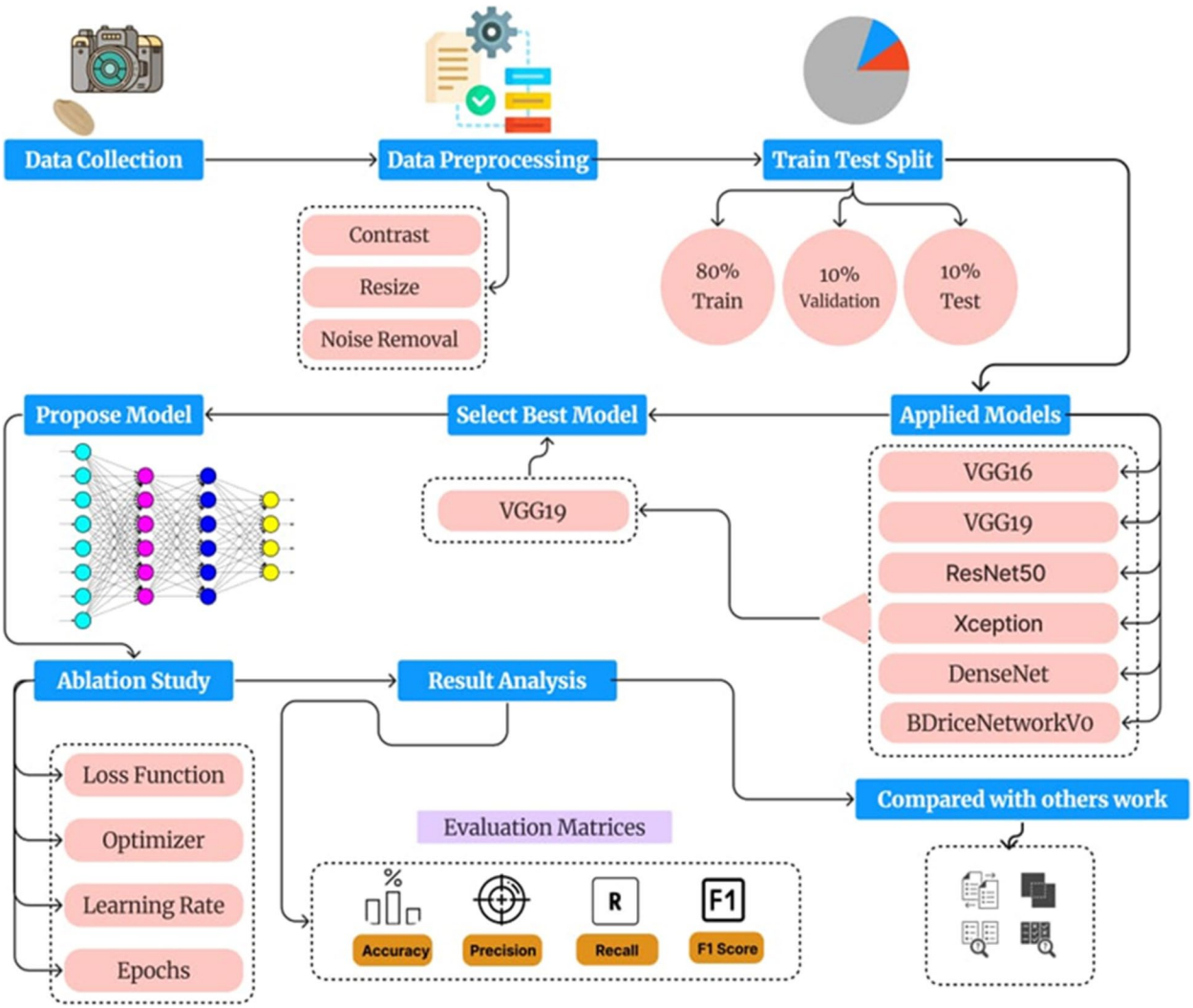


Fig. 2 End-to-end framework for preprocessing and classification of rice varieties using DL models

multiple regions. To improve data quality and uniformity, all images are subjected to a variety of preprocessing techniques, including noise reduction, resizing all images to 224 × 224, and enhancing contrast, with the aim of amplifying distinctive visual traits relevant for classification. Following preprocessing, the dataset is divided into three distinct subsets: 80% for training, 10% for validation, and 10% for testing. The dataset was shuffled prior to splitting to prevent order bias. Training utilized an early stopping mechanism monitoring the validation loss with a patience of 10 epochs to prevent overfitting. This ensures a balanced distribution while preventing data leakage, thereby supporting both effective model training and robust performance evaluation. To minimize overfitting and enhance the model's ability to generalize to unseen data, data augmentation techniques such as rotation (15° clock/anti-clock wise), zoom (150%), and horizontal/vertical flipping are applied to the training set. A fixed random seed (42) was used for data splitting to ensure reproducibility. The model implementation utilizes various advanced deep CNNs, including VGG16, VGG19, ResNet50, Xception, and DenseNet. In addition to these pretrained architectures, a custom-designed CNN model, named BDriceNetworkV0, is developed and trained from scratch to serve as a benchmark for comparison. Each model is optimized and fine-tuned using the same dataset and evaluated under identical conditions to ensure fairness in performance assessment. The models are evaluated using standard performance metrics, including accuracy, precision, recall, and F1-score, to assess their effectiveness. Comparative analysis with results from prior studies further contextualizes the improvements achieved by the proposed models.

Network architectures

VGG19

The VGG network (Fig. 3), developed by Oxford's Visual Geometry Group, is a deep CNN aimed at improving image classification through increased depth and simplicity [16]. VGG19, with 19 layers including 16 convolutional and 3 fully connected layers, uses 3×3 filters to capture complex features. It processes 224×224×3 images, resized and normalized. The architecture has five convolutional blocks, each followed by max-pooling to reduce spatial size. Blocks 1 and 2 have 2 layers with 64 and 128 filters; blocks 3 to 5 have 4 layers with 256, 512, and 512 filters, down to 7×7. The output is flattened and passed through two 4 096-neuron fully connected layers, then a final 1 000-neuron softmax layer. ReLU activation follows each layer, with dropout preventing overfitting. The model is trained using cross-entropy loss and optimized with stochastic gradient descent (SGD) and momentum [17]. Pretrained on ImageNet, it is widely fine-tuned for specific tasks and shows strong generalization.

ResNet50

ResNet50 is a deep CNN that solves the vanishing gradient problem with residual learning and skip connections [18]. Unlike VGG19, it uses skip connections to train its 50-layer structure effectively (Fig. 4). It processes 224×224×3 images through a 7×7 stride-2 convolution, then max-pooling. Its architecture includes four residual blocks with 3, 4, 6, and 3 units. Each unit has 3 convolutional layers with batch normalization, ReLU, and shortcuts. ResNet-50 has about 25 million parameters, making it more parameter-efficient than VGG19 while performing better. It uses cross-entropy loss and SGD with momentum. Trained on ImageNet, ResNet-50 is popular for transfer learning and computer vision, balancing accuracy and efficiency.

DenseNet

DenseNet is a CNN that improves feature propagation by connecting each layer to the next, unlike ResNet's skip connections. It processes 224×224×3 images through an initial convolution and 4 dense blocks [19], each with multiple dense layers that receive features from all previous layers (Fig. 5). Dense layers typically include batch normalization, ReLU, and 3×3 convolutions. Transition layers with 1×1 convolutions and 2×2 pooling reduce spatial size and feature maps between blocks. For ImageNet, it uses global average pooling before the final 1 000-class layer. DenseNet's main advantage is feature reuse, offering better gradient flow and fewer parameters, which enhances performance and reduces computational

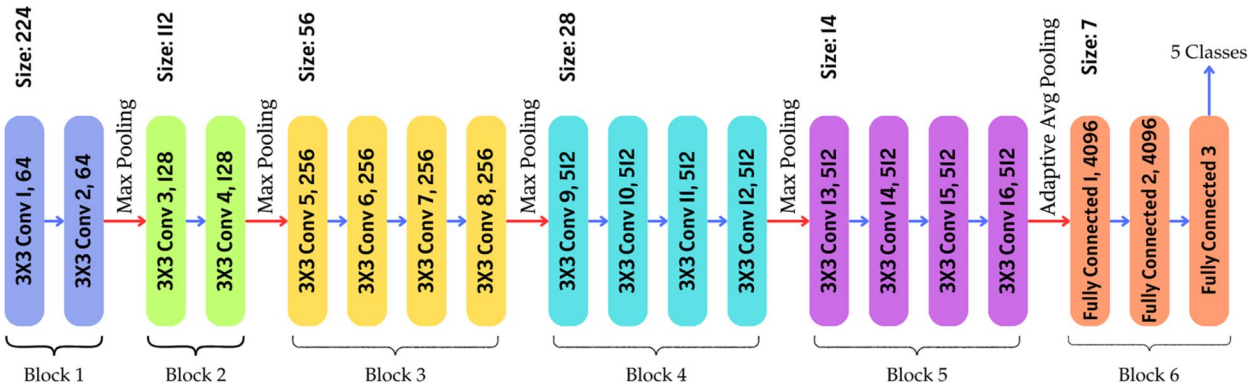


Fig. 3 Network architecture of the VGG19 model

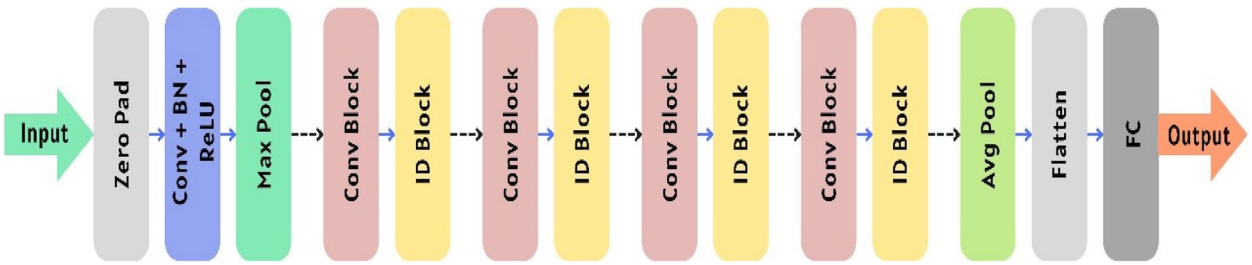


Fig. 4 Network architecture of the ResNet50 model

load. It achieves high accuracy using cross-entropy loss and SGD, and pretrained models excel in transfer learning, outperforming ResNet and VGG with less computation.

VGG16

VGG16 is a robust CNN used for image classification [20]. It has 16 layers: 13 convolutional and 3 fully connected. It uses 3×3 filters for feature extraction (Fig. 6). The model processes 224×224×3 images through 5 blocks. The first 2 use 64 and 128 filters with max-pooling; the 3rd and 4th have 256 and 512 filters; the 5th contains 512 filters. Max-pooling halves spatial dimensions. Features are flattened after convolution and pass through 3 fully connected layers: 2 with 4 096 neurons each, and a 1 000-neuron output for ImageNet. ReLU activation and dropout reduce overfitting. VGG16, optimized with cross-entropy loss and momentum SGD, has about 138 million parameters, making it computationally intensive but effective.

Xception

Xception is a deep CNN that improves the inception architecture by replacing traditional modules with depthwise separable convolutions [21]. It processes 224×224×3 images in three stages: entrance, middle, and exit flow (Fig. 7). The entrance flow reduces dimensions from 112×112 to 7×7 using convolutional layers and residual blocks. The middle flow has 8 residual blocks, maintaining 7×7 size while increasing feature depth. The exit flow includes 2 residual blocks and global average pooling. It employs batch normalization, ReLU, and skip connections for efficient deep network training. Pretrained on ImageNet, it excels in transfer learning for classification, recognition, and segmentation.

BDriceNetworkV0

In this study, we introduced BDriceNetworkV0, a custom-designed CNN tailored specifically for the classification of Bangladeshi rice varieties. BDriceNetworkV0 was designed with a shallow architecture (3 convolutional blocks) to minimize computational cost while retaining sufficient capacity for feature extraction. The filter progression (32 → 64 → 128) allows for hierarchical learning of increasingly complex features without the parameter explosion seen in deeper networks. The model processes input images with dimensions 224×224×3, which are then passed through several convolutional, pooling, and

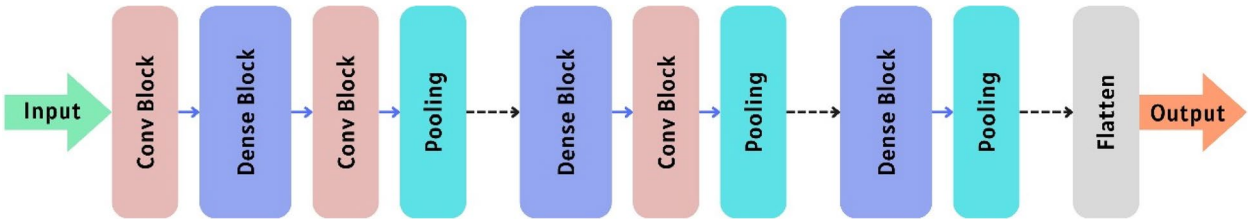


Fig. 5 Network architecture of the DenseNet model

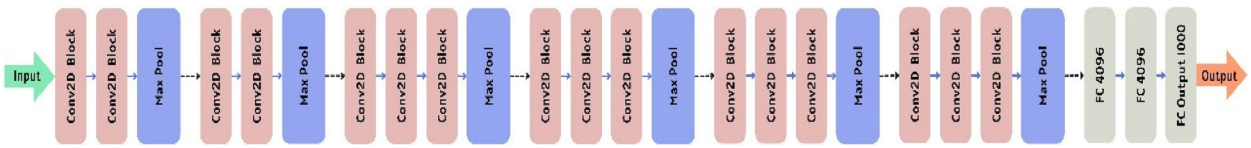


Fig. 6 Network architecture of the VGG16 model

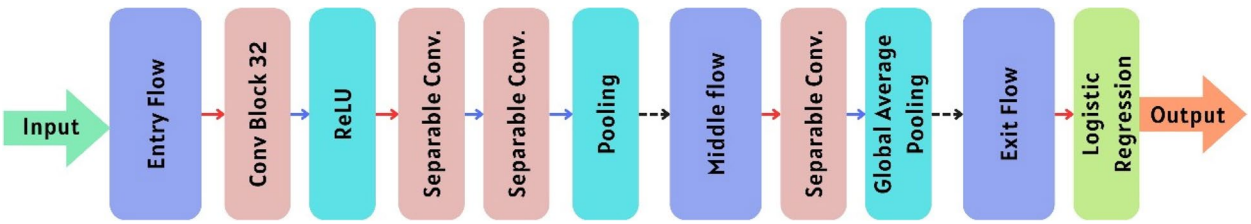


Fig. 7 Network architecture of the Xception model

fully connected layers to extract hierarchical features (Fig. 8). The architecture begins with a 3×3 convolutional layer with 32 filters, followed by ReLU activation and max-pooling to reduce the spatial dimensions. This process is repeated in subsequent layers: a 2nd convolutional layer with 64 filters and a 3rd with 128 filters. Each convolution is followed by ReLU activation and max-pooling, gradually reducing the spatial dimensions of the feature maps from 224×224 to 56×56, 28×28, and ultimately 14×14. This structure ensures that relevant features are extracted at varying levels of abstraction while minimizing the spatial resolution of the input. After the convolutional layers, the feature maps are flattened into a 1D vector and passed through 2 fully connected layers: one with 1 024 neurons and the second with 512 neurons. Both layers use ReLU activation to introduce nonlinearity, enhancing the model's ability to learn complex patterns. To mitigate the risk of overfitting, a dropout rate of 0.5 is applied between these fully connected layers. The final layer of the network is a softmax output layer with 5 neurons, corresponding to the 5 rice varieties under classification: Basmati, Chinigura, Jirashail, Kataribhog, and Paijam. The network is trained using cross-entropy loss, which is standard for classification tasks, and ReLU activations are applied throughout to further introduce nonlinearity and enable effective learning. BDriceNetworkV0 strikes a careful balance between depth and computational efficiency, making it particularly well-suited for fine-grained image classification tasks like rice variety identification. Its design ensures robust performance, leveraging the strengths of convolutional layers for feature extraction and fully connected layers for classification.

Evaluation metrics

To evaluate the performance of the proposed image-based classification system for Bangladeshi rice varieties, several key metrics were employed. The experimental outcomes were categorized into four distinct classes, each representing a different type of prediction: True positive (TP) refers to the accurate detection of a rice grain that is correctly identified and classified into its respective variety; False positive (FP) occurs when a rice grain is incorrectly classified into a category that it does not belong to, resulting in a misclassification; True negative (TN) represents the correct identification of a negative class, where a non-rice object or incorrect rice variety is rightly predicted as not belonging to the target category; False negative (FN) indicates a missed detection, where a rice grain belonging to a specific class is not identified, leading to an undetected or misclassified rice variety [22, 23]. Metrics were calculated for each class in a one-*vs.*-all manner and then averaged (macro-average). TP represents correctly classified grains of the target class, while FP represents grains of other classes incorrectly labeled as the target class. These outcomes are essential for calculating various performance metrics that reflect the accuracy and effectiveness of the classification model. The key statistical measures used in the evaluation process are derived from the following equations which provide a standardized approach for assessing model performance in image classification tasks.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{1}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$

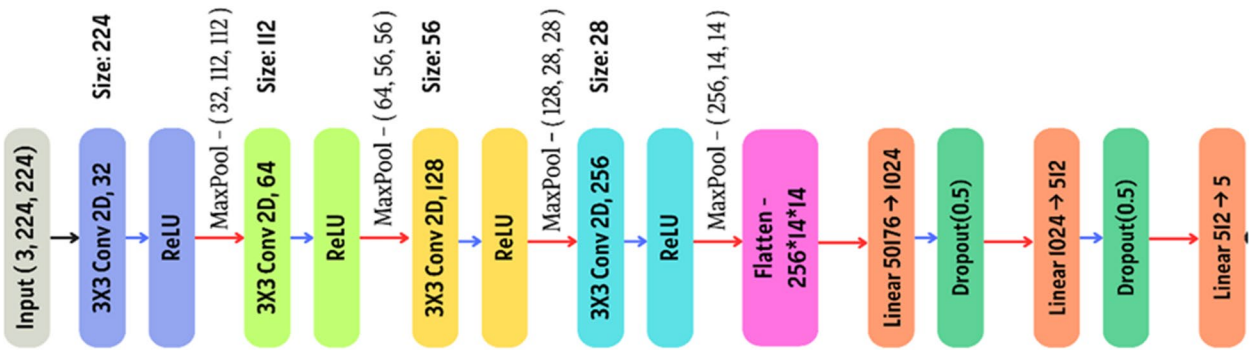


Fig. 8 Custom designed BDriceNetworkV0 model architecture for rice variety classification

$$F1-\text{score} = 2/(\left(\frac{1}{\text{Precision}}\right) + \left(\frac{1}{\text{Recall}}\right)) \quad (4)$$

Results and discussion

Training dynamics and convergence

The training phase of the study was carefully designed to ensure optimal learning conditions for each deep CNN evaluated. The selected models were trained under consistent parameters, with adjustments made to accommodate individual architectural characteristics. A summary of the hyperparameter settings used for each model is presented in Table 2. All models were trained using a batch size of 32 and optimized with the Adam optimizer, known for its adaptive learning rate and computational efficiency in large-scale DL tasks [24]. To balance stability and convergence speed, the learning rate was set to 0.000 001 for VGG16, VGG19, and DenseNet, 0.000 01 for ResNet50 and Xception, and 0.000 1 for BDriceNetworkV0. The relatively higher learning rate for BDriceNetworkV0 was chosen to accelerate convergence given its simpler architecture compared to the deeper pretrained networks. A similar approach to hyperparameter tuning has been demonstrated in other studies. For instance, Latif et al. [25] proposed a modified VGG19-based transfer learning approach, attaining 96.08% accuracy for 6 rice leaf disease classes. Additionally, the decision to train VGG16 for 21 epochs while DenseNet required 40 epochs is consistent with findings in the literature that suggest deeper networks tend to require more epochs to converge due to their higher complexity [26]. This observation highlights the importance of adjusting training durations according to model architecture.

The number of training epochs varied across models based on convergence behavior during validation. VGG16 required 21 epochs, whereas DenseNet required 40 epochs, suggesting that deeper networks require more iterations for stable learning. The custom model BDriceNetworkV0 was trained for 29 epochs, balancing training duration with generalization performance. The relatively fewer epochs required for VGG16 in this study can be attributed to its shallow architecture compared to DenseNet. This finding supports the assertion made by Qassim et al. [27] that VGG16 tend to achieve quicker convergence without the need for excessive training time. However, these models may suffer from limitations in feature representation compared with more complex networks such as DenseNet or ResNet, which incorporate deeper layers and feature reuse, thereby enhancing their ability to capture intricate patterns in the data. This research highlights the benefits of employing DL models, namely CNN, which autonomously extract intricate features from raw image data, in contrast to studies that have concentrated on classifying rice types through conventional ML methods [28]. Pillai et al. [7] observe that DL models, particularly CNNs, have transformed image classification tasks by effectively modeling spatial hierarchies, which is essential for distinguishing subtle visual differences among agricultural products such as rice varieties.

Table 2 Training epochs and hyperparameter settings for deep CNN models used in rice variety classification

Model	Epoch	Batch Size	Learning Rate	Optimizer
VGG16	21	32	0.000 001	Adam
VGG19	26	32	0.000 001	Adam
ResNet50	30	32	0.000 01	Adam
Xception	26	32	0.000 01	Adam
DenseNet	40	32	0.000 001	Adam
BDriceNetworkV0	29	32	0.000 1	Adam

Comparative model performance

The performance evaluation across the selected CNN architectures reveals distinct variations in generalization capability and convergence behavior. Notably, VGG19 demonstrated the most consistent and superior performance with both the lowest training (0.019 0) and validation loss (0.014 2), accompanied by the highest validation accuracy (99.40%) (Fig. 9). This indicates a robust learning curve with minimal overfitting, as supported by its tightly aligned loss and accuracy curves. Although VGG16 achieved slightly lower validation accuracy (99.07%), its performance remained stable, with modest loss values, reinforcing its suitability for lightweight applications. On the other hand, ResNet50, while popular for its residual learning capability, yielded a relatively lower validation accuracy (98.67%) and the highest validation loss (0.044 5) among the pretrained models, suggesting underfitting or less compatibility with the dataset's specific features. Interestingly, DenseNet and Xception achieved identical validation accuracies (99.20%), yet DenseNet exhibited a higher training loss, implying a more gradual learning process. The custom BDriceNetworkV0, although it did not outperform pretrained models, achieved a competitive validation accuracy (98.44%) with reasonable convergence. Its sharper early-stage accuracy improvements, followed by plateauing curves, highlight its potential as a computationally efficient alternative when fewer resources are available. The absence of significant divergence between training and validation curves across all models suggests that overfitting was effectively mitigated, likely due to balanced data preprocessing, regularization, and early stopping strategies. However, the trade-off between training duration, model complexity, and

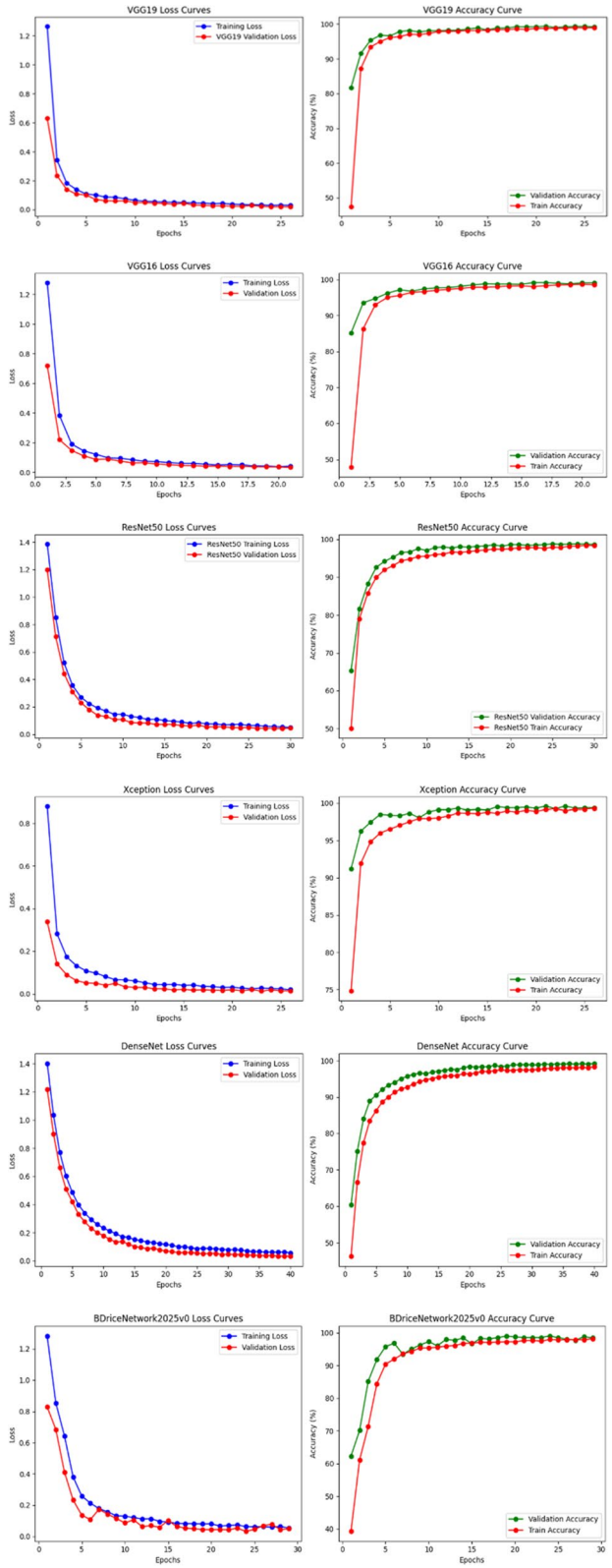


Fig. 9 Training-validation loss and training-validation accuracy curves for CNN models on Bangladeshi rice variety classification

performance highlights the importance of task-specific model selection over brute computational power.

The performance trend observed in this study aligns with existing literature where VGG-based architectures have proven highly effective in agricultural image classification due to their structured feature extraction and simplicity [29]. VGG architectures have also proven valuable in texture-based leaf disease classification, outperforming traditional methods in processing speed and discriminative power [30]. VGG19 has shown superior performance in accurately categorizing diverse rice varieties, albeit with increased computational demands [31]. Although ResNet architectures have demonstrated success in complex image domains like medical imaging [32], their lower performance here may reflect an overcomplexity for a relatively simpler texture-based classification problem. The observation that DenseNet achieved high accuracy is supported by Kamilaris et al. [33], who reported its strength in capturing fine-grained visual features in agricultural datasets through dense connectivity. Furthermore, the performance of the custom BDriceNetworkV0 model demonstrates the potential of tailored lightweight architectures, particularly in resource-constrained environments.

Analysis of the best model (VGG19)

The evaluation results, presented in Tables 3 and 4, provide significant insights into the relative effectiveness of each model for accuracy and loss measures. VGG19 distinguished itself as the leading model, achieving the

Table 3 Test performance of DL models

Model	Test accuracy (%)	Test loss
VGG16	98.73	0.033 9
VGG19	99.73	0.013 1
ResNet50	98.67	0.040 5
DenseNet	98.20	0.049 9
Xception	99.07	0.028 1
BDriceNetworkV0	98.60	0.049 1

Table 4 Overall performance outcomes of the DL models

Model	Accuracy (%)	Precision	Recall	F1 score
VGG16	98.73	0.984 7	0.984 7	0.984 6
VGG19	99.73	0.997 3	0.997 3	0.997 3
ResNet50	98.67	0.986 1	0.986 0	0.986 0
DenseNet	98.20	0.982 2	0.982 0	0.982 0
Xception	99.07	0.986 0	0.986 0	0.986 0
BDriceNetworkV0	98.60	0.985 4	0.985 3	0.985 3

highest test accuracy of 99.73% and the lowest test loss of 0.013 1, demonstrating its strong generalization ability. This superior performance is further underscored by VGG19's validation metrics, which also surpassed other models, demonstrating its reliability and robustness for this classification task. Xception, another strong contender, achieved a test accuracy of 99.07% and a test loss of 0.028 1. Although slightly less accurate than VGG19, Xception demonstrated commendable performance and was highly competitive among the models. VGG19's outstanding performance in this study is consistent with findings from other studies that demonstrate the effectiveness of pre-trained models such as VGG19 for plant species and crop variety classification. Fine-tuning these models has shown significant improvements in classification accuracy, with VGG16 achieving over 99.96% accuracy after fine-tuning [34]. VGG16 followed closely behind, achieving a test accuracy of 98.73% and a test loss of 0.033 9. ResNet50 showed a comparable performance with a test accuracy of 98.67%, though it had a slightly higher test loss of 0.040 5. Both models performed well, with VGG16 showing slightly better results in terms of accuracy and ResNet50 offering a competitive edge in terms of precision and recall. DenseNet and the custom model BDriceNetworkV0 demonstrated lower performance metrics, with test accuracies of 98.20% and 98.60%, respectively. In some case, custom hybrid models can outperform standard architectures like VGG16 and MobileNetV2, achieving 99% accuracy [35]. While custom models like BDriceNetworkV0 and DenseNet did not perform as well as VGG19, their relatively lower accuracies are not uncommon in the field of DL for agricultural classification. DenseNet's test accuracy of 98.20% and BDriceNetworkV0's performance (98.60%) indicate that custom networks, although potentially promising, often require further optimization to achieve performance comparable to well-established architectures.

In terms of classification quality, VGG19 consistently excelled across all performance metrics. With an average precision of 0.997 3, VGG19 proved to be the most reliable model for this task, ensuring minimal FPs and FNs. These metrics not only reinforce its high accuracy but also confirm its superior ability to balance precision and recall, providing an optimal solution for rice variety classification. The high precision and recall values observed for VGG19 reflect its strong ability to minimize FPs and FNs, making it ideal for tasks where classification accuracy is critical, such as in agricultural monitoring. Studies by Miftahushudur et al. [36] have highlighted the importance of F1-score in agricultural image classification, suggesting that models with balanced precision and recall, such as VGG19, offer superior performance in real-world applications. This is further supported by research on specific crops, such as maize [37], where F1-scores are used to evaluate model effectiveness.

Confusion matrix

To better assess the classification robustness of the models, confusion matrices were created for all six architectures tested: VGG16, VGG19, ResNet50, Xception, DenseNet, and the custom BDriceNetworkV0 (Fig. 10). These matrices offer a detailed view of prediction performance, highlighting particular misclassification patterns between rice varieties that simple accuracy metrics might miss. This matrix is a 5×5 representation, with one axis showing actual (true) values and the other showing predicted values by the model. In all cases the diagonal cells contain substantially larger values than the off-diagonal elements, indicating that every network makes the majority of its predictions correctly. The VGG19 model (Fig. 10A) showed the most robust performance, consistent with its superior overall accuracy of 99.73%. The matrix shows a nearly perfect diagonal, indicating that the model correctly predicted 299 or 300 grains for each class. Only 2 misclassifications were observed across the entire test set: a single Basmati grain was incorrectly labeled as Jirashail, and 1 Kataribhog grain was mislabeled as Paijam. This confirms that VGG19 effectively captures subtle morphological distinctions between visually similar long-grain varieties. The Xception model (Fig. 10D), which achieved the second-highest accuracy (99.07%), also demonstrated exceptional separability. The off-diagonal elements were minimal, with rare confusions occurring primarily between Basmati and Jirashail. This suggests that Xception's depthwise separable convolutions are highly effective at extracting discriminative features, though slightly less precise than VGG19 in resolving the finest textural details of the grain surfaces. VGG16 (Fig. 10B) and ResNet50 (Fig. 10C) displayed comparable performance tiers (98.73% and 98.67%, respectively). While VGG16 maintained strong diagonal dominance, ResNet50 showed slightly higher confusion in specific categories. For instance, ResNet50 mislabeled a small cluster of Jirashail grains as Basmati and Kataribhog grains as Chinigura. These errors are attributable to the high degree of visual similarity in grain length and opacity between these specific varieties, indicating that ResNet50's residual features, while robust, occasionally struggle to differentiate the boundaries of highly similar classes compared to the VGG architectures in this specific domain. The DenseNet model (Fig. 10E), despite its efficient feature reuse, yielded the lowest accuracy among the tested models (98.20%). The confusion matrix reveals a higher frequency of

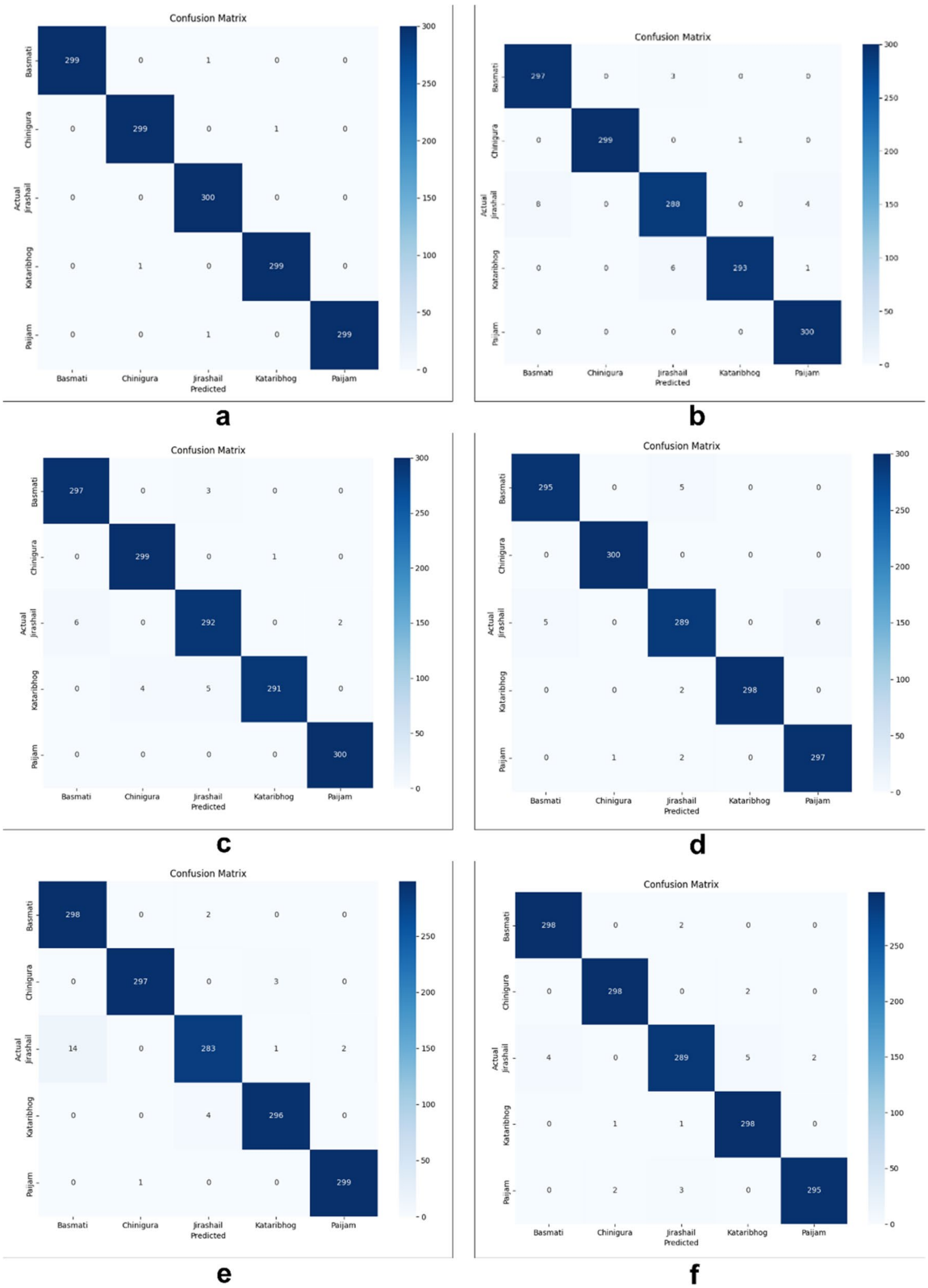


Fig. 10 Confusion matrices. **A**. VGG19, (**B**). VGG16, (**C**). Resnet50, (**D**). Xception, (**E**). DenseNet, and (**F**). BDriceNetworkV0

off-diagonal dispersals, particularly distinguishing between Paijam and other medium-grain varieties. This suggests that while DenseNet is computationally efficient, its feature maps may be less distinct for the subtle inter-class variations present in rice morphological analysis compared to deeper or wider architectures.

Finally, the custom BDriceNetworkV0 (Fig. 10F) demonstrated impressive consistency for a lightweight model, achieving an accuracy of 98.60%. Its confusion matrix closely mirrors that of ResNet50, with misclassifications largely confined to the most structurally similar variety pairs (e.g., Basmati *vs*. Jirashail). This result highlights the effectiveness of the custom architecture, proving that a streamlined model designed specifically for this domain can achieve classification reliability comparable to much larger, pre-trained networks like ResNet50. These patterns show that most errors occur among morphologically similar varieties—particularly between the slender-grained Basmati and Jirashail, or between Jurashail and Kataribhog—where subtle differences in grain length, width or texture are hard to capture. The confusion-matrix analysis indicates that, although all 6 deep networks achieve high overall accuracy ($\geq$ 97%), misclassifications arise mainly among morphologically similar varieties. Similar observations have been reported by Tran-Thi-Kim et al. [38].

Our findings corroborate the broader literature on rice-variety classification. Khin et al. [39] applied a diffusion CNN to classify five Myanmar rice seed types and reported that the confusion matrix contained predominantly diagonal elements, indicating a very low rate of misclassification and near-perfect accuracy. The authors emphasized that the across-diagonal elements were significantly higher than the off-diagonal entries and concluded that DCNNs can achieve approximately 99% accuracy. Similar trends were observed in our VGG19 results, suggesting that deeper convolutional architectures can generalize well across different rice datasets when sufficient training data are available. Several recent studies have also highlighted that misclassification occurs mainly between varieties with similar shapes or textures. Asif et al. [40] trained a CNN on 75 000 images representing 5 varieties (Arborio, Basmati, Ipsala, Jasmine and Karacadag). They observed that their model correctly classified 1 478 Arborio, 1 483 Basmati and 1 495 Ipsala grains, yet misclassified 16 Ipsala grains as Basmati and 10 Jasmine grains as Karacadag. The authors noted that misclassification was common among closely related classes such as Basmati and Ipsala and suggested that data augmentation and feature engineering could reduce ambiguity. This mirrors the patterns seen in our models, where Jirashail and Basmati grains were occasionally confused.

Performance analysis and visualization of the best model

Table 5 displays the classification performance of the VGG19 model, which is the best performing model in this study. The model's capability to categorize 5 Bangladeshi rice types was evaluated through precision, recall, and F1-score metrics. The VGG19 model attained excellent scores in precision, recall, and F1-score for 4 rice varieties: Basmati, Chinigura, Kataribhog, and Paijam (Table 5). The findings reveal that the model accurately classified these kinds without any FP or FN, demonstrating outstanding performance. For the Jirashail class, the model recorded a slight reduction in precision (0.993 3), but it still achieved perfect recall (1.00) and F1-Score (0.996 6), signifying that the model had only a minimal trade-off between precision and recall while maintaining high overall performance. The overall accuracy, macro-average, and weighted-average scores further underscores the model's consistent and balanced performance across all rice varieties (Table 5).

The ROC curve in Fig. 11 provides further evidence of the model's outstanding performance. The curves for the 5 rice varieties are nearly indistinguishable from each other and converge toward the ideal top-left corner of the plot. This perfect alignment suggests that the model achieved an ideal classification rate for each variety, maintaining high true positive rate (TPR) and low false positive rate (FPR). The area under the curve (AUC) scores for all rice varieties were reported as 1.00, further solidifying the model's ability to perfectly separate the rice classes with no loss in sensitivity or specificity. The steep upward trend of the curves in the ROC plot further reflects the model's capacity to maintain a low FPR while ensuring a high TPR for each class. The excellent AUC scores result from exceptional class separability under the experimental conditions. We attribute this to several factors: 1) the controlled imaging setup minimized noise

Table 5 Classification performance of the VGG19 model on Bangladeshi rice varieties

Class	Precision	Recall	F1-score
Basmati	1	0.996 6	0.998 3
Chinigura	0.996 6	0.996 6	0.996 6
Jirashail	0.993 3	1	0.996 6
Kataribhog	0.996 6	0.996 6	0.996 6
Paijam	1	0.996 6	0.998 3
macro avg	0.997 3	0.997 3	0.997 3
weighted avg	0.997 3	0.997 3	0.997 3

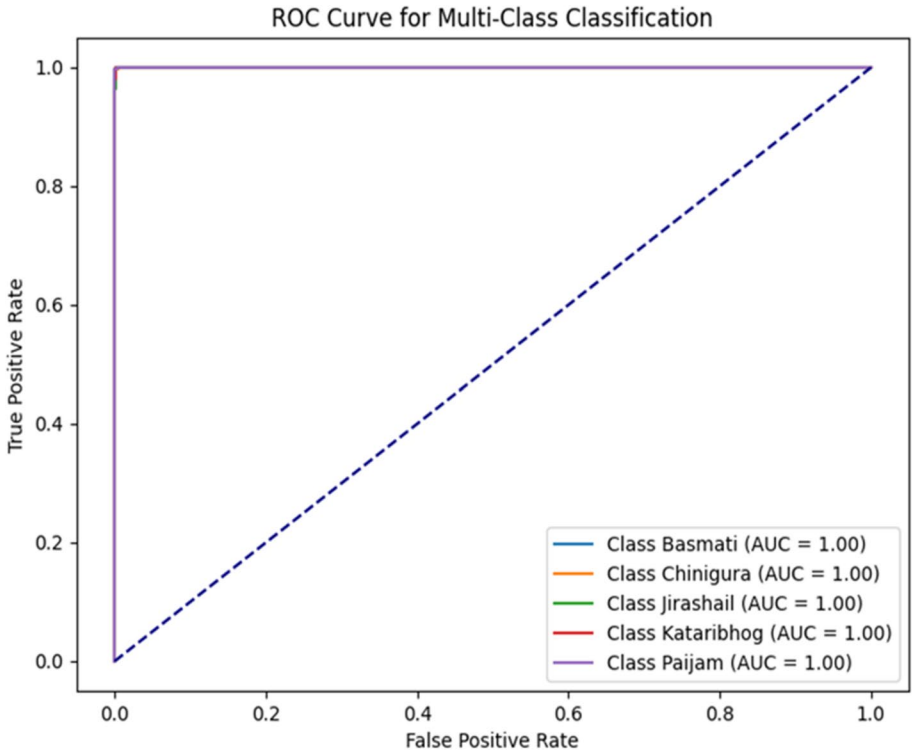


Fig. 11 ROC curve for multi-class classification of Bangladeshi rice varieties

and intra-class variance, 2) the 5 selected varieties have distinctly different morphological traits (as described in Sect. "Dataset details"), and 3) the dataset was carefully curated and balanced. We verified our data splitting procedure to ensure no cross-contamination between training and test sets. While this performance reflects ideal laboratory conditions, it sets a strong upper benchmark for this specific classification task. An example of high-performance classification is provided by LaBreche et al. [41], who created a breast tumor predictor with an AUC of 0.97, achieving 89% sensitivity and 100% specificity. This strong performance in the ROC analysis complements the high precision, recall, and F1-score values, reinforcing the VGG19 model as a reliable classifier for Bangladeshi rice varieties. The ability of VGG19 to achieve such high AUC scores across all rice varieties indicates its robustness and superior performance in multi-class classification tasks.

The visual results (Fig. 12) validate the quantitative metrics presented earlier, reinforcing the exceptional performance of VGG19 in this task. The correct identification of these rice varieties, even for potentially challenging cases, demonstrates the model's robustness. Moreover, the consistency of predictions across different rice varieties further emphasizes the model's ability to generalize effectively, an essential factor in real-world applications such as rice variety classification in agricultural fields.



Fig. 12 Classification examples for VGG19 for 5 Bangladeshi rice varieties

Limitations

This study's dataset is confined to only 5 rice varieties, thereby failing to encompass the complete spectrum of rice diversity present in Bangladesh. Augmenting the dataset to incorporate a greater diversity of types and supplementary environmental variables may enhance the models' generalizability. The rice grain images were obtained in a controlled laboratory environment with uniform lighting and background. This may restrict the applicability of the models to field settings or real-world contexts characterized by noise and variability. This study relies solely on visual data. Including additional features such as aroma, grain weight, or spectral data could enhance classification performance and robustness. Lastly, the computational cost of training deep models, particularly VGG19 and DenseNet, poses a challenge for real-time deployment in resource-limited settings, suggesting the need for optimization or lighter architectures for practical applications. While VGG19 is accurate, it is heavy; therefore, the Custom Model (BDriceNetworkV0) is the solution for the "Computational Cost" limitation.

Conclusion and future works

This research successfully demonstrates the application of DL, specifically CNNs, for the automated classification of Bangladeshi rice varieties. The findings indicate that pre-trained models, particularly VGG19, exhibit great accuracy and generalizability, rendering them appropriate for agricultural applications like quality control and fraud prevention. The custom-designed model, BDriceNetworkV0, offers a viable solution for low-resource environments; however, further optimization is required to achieve improved performance. Future research may concentrate on augmenting the dataset to encompass a wider array of rice varieties and environmental variables, alongside investigating the incorporation of other sensory data (such as aroma and texture) to improve classification precision. Furthermore, addressing the computational challenges of DL models by developing more efficient architectures or leveraging transfer learning could facilitate real-time, scalable applications in agricultural settings. Exploring hybrid models combining CNNs with other ML techniques may also provide new avenues for improving classification accuracy and robustness. Ultimately, this study paves the way for the broader adoption of AI-driven solutions in precision agriculture, supporting efforts to enhance food security and supply chain integrity.

Acknowledgements

The author thanks and acknowledges the laboratory of the Department of Computer Science and Engineering at Sylhet Engineering College (SEC), Sylhet 3100, Bangladesh, for granting access to the facilities during the research.

Authors' contributions

Syme Al-Azad Shuvo: Writing-original draft, visualization, methodology, software, data collection; Khalilur Rahman, Abdullah Al Naseeh Chowdhury, and Gourab Roy: Methodology, software, data collection; Nafiz Nahid and Mahmud Hasan: Writing-original draft, visualization, software; Md. Abu Naser Mojumder: Writing-original draft, writing-reviewing and editing, validation, methodology, supervision; Md. Janibul Alam Soeb: Writing-original draft, writing-reviewing and editing, data collection; Md. Fahad Jubayer: Writing-original draft, writing- reviewing and editing, validation, methodology, software, supervision.

Data availability

Data will be made available upon reasonable request to the corresponding author.

Declarations

Competing interests

The authors declare no competing interests.

Received: 26 September 2025 Revised: 7 January 2026 Accepted: 8 January 2026
Published online: 19 January 2026

References

1. Farahnakian F, Sheikh J, Farahnakian F, et al. A comparative study of state-of-the-art deep learning architectures for rice grain classification. J Agric Food Res. 2024;15:100890. https://doi.org/10.1016/j.jafr.2023.100890.
2. Islam MM, Himel GMS, Moazzam MG, et al. Rice variety classification using next generation convolutional networks. J Eng. 2025;1:e70102. https://doi.org/10.1049/tje2.70102.
3. Fukagawa NK, Ziska LH. Rice: importance for global nutrition. J Nutr Sci Vitaminol. 2019;65:S2–3. https://doi.org/10.3177/jnsv.65.S2.
4. Mamun MAA, Nihad SAI, Sarker MR, et al. Spatiotemporal mapping of rice acreage and productivity growth in Bangladesh. PLoS ONE. 2024;19(3):e0300648. https://doi.org/10.1371/journal.pone.0300648.
5. Hanafiah NM, Mispan MS, Lim PE, et al. The 21st century agriculture: when rice research draws attention to climate variability and how weedy rice and underutilized grains come in handy. Plants. 2020;9(3):365. https://doi.org/10.3390/plants9030365.
6. Arora B, Bhagat N, Saritha LR, et al. Rice grain classification using image processing & machine learning techniques. IEEE Int Conf Invent Comput Technol. 2020;2020:205–8.
7. Pillai R, Sharma N, Chauhan R, et al. CNN-based approach for classification of rice varieties. IEEE Int Conf Futur Technol. 2023;2023:1–5. https://doi.org/10.1109/INCOFT60753.2023.10425548.
8. Shamim MI, Pal B, Arora AS, et al. A deep convolutional neural network approach to rice grain purity analysis. Comput Analy Network: Proceedings. 2020:179–89. https://doi.org/10.1007/978-981-15-2414-1_19.
9. Zheng K, Sun M. Rice quality identification based on transfer learning. IEEE Int Conf Comput Info Processing Adv Edu. 2024;2024:385–9. https://doi.org/10.1109/CIPAE64326.2024.00075.
10. Kelishami AS, Kelishami SS. Cell phone image-based persian rice detection and classification using deep learning techniques. arXiv Preprint. arXiv:2404;13555. https://doi.org/10.48550/arXiv.2404.13555.
11. Sharma K, Sethi GK, Bawa RK. A comparative analysis of deep learning and deep transfer learning approaches for identification of rice varieties. Multi Tool Appli. 2025;84(10):6825–42. https://doi.org/10.1007/s11042-024-19126-7.
12. Razavi M, Mavaddati S, Koohi H. Resnet deep models and transfer learning technique for classification and quality detection of rice cultivars. Expt Sys Appli. 2024;247:123276. https://doi.org/10.1016/j.eswa.2024.123276.
13. Priya PK, Kirupa P, Thilakaveni P, et al. DeepRiceTransfer: exploiting CNN transfer learning for effective rice variety classification. IEEE Int Conf Soci Sust Inno Technol Eng. 2024;2024:102–7. https://doi.org/10.1109/SASITE58663.2024.00025.

14. Ahad MT, Li Y, Song B, et al. Comparison of CNN-based deep learning architectures for rice diseases classification. Artificial Intelligence in Agriculture. 2023;9:22–35. https://doi.org/10.1016/j.aiia.2023.07.001.
15. Chen J, Chen J, Zhang D, et al. Using deep transfer learning for image-based plant disease identification. Comput Electron Agric. 2020;173:105393. https://doi.org/10.1016/j.compag.2020.105393.
16. Simonyan K, Zisserman A. Very deep convolutional networks for large-scale image recognition. arXiv Preprint. arXiv:2014;1409:1556. https://doi.org/10.48550/arXiv.1409.1556.
17. Kumar HBB, Chennamma HR. Classification of computer graphic images using deep learning. Int Conf Adv Electri Comput Technol. 2020;2020:137–42. https://doi.org/10.1007/978-981-15-9019-1_12.
18. Zhao CH, Leong WY. Optimization solutions and simple innovative solution research on ResNet50 model. ASM Sci J. 2025;20(1). https://doi.org/10.32802/asmscj.2025.2002.
19. Huang G, Liu Z, Maaten LVD, et al. Densely connected convolutional networks. 2017 IEEE Int Conf Comput Vision Pat Recog. 2017:4700–8. https://doi.org/10.48550/arXiv.1608.06993.
20. Yang R. Convolutional neural network for image classification research-based on VGG16. IEEE Int Conf Imag Proces Comput Vision Machine Learning. 2024;2024:213–7. https://doi.org/10.1109/ICICML63543.2024.10958042.
21. Chollet F. Xception: deep learning with depthwise separable convolutions. 2017 IEEE Int Conf Comput Vision Pattern Recog. 2017:1251–8. https://doi.org/10.48550/arXiv.1610.02357.
22. Jubayer MF, Ruhad FM, Kayshar MS, et al. Detection and identification of honey pollens by YOLOv7: a novel framework toward honey authenticity. ACS Agric Sci Technol. 2024;4(7):747–58. https://doi.org/10.1021/acsagscitech.4c00220.
23. Soeb MJA, Jubayer MF, Tarin TA, et al. Tea leaf disease detection and identification based on YOLOv7 (YOLO-T). Sci Rep. 2023;13(1):6078. https://doi.org/10.1038/s41598-023-33270-4.
24. Kingma DP, Ba J. Adam: A method for stochastic optimization. arXiv Preprint. arXiv:2014;1412:6980. https://doi.org/10.48550/arXiv.1412.6980.
25. Latif G, Abdelhamid SE, Mallouhy RE, et al. Deep learning utilization in agriculture: detection of rice plant diseases using an improved CNN model. Plants. 2022;11(17):2230. https://doi.org/10.3390/plants11172230.
26. Chen Y, Yuille A, Zhou Z. Which layer is learning faster? a systematic exploration of layer-wise convergence rate for deep neural networks. 11th Int Conf Learning Represent. 2023. https://openreview.net/forum?id=wlMDF1jQF86. Accessed on: 12 July 2025.
27. Qassim H, Verma, Feinzimer D. Compressed residual-VGG16 CNN model for big data places image recognition. 2018 IEEE 8th Annual Comput Commun Workshop Conf. 2018:169–75. https://doi.org/10.1109/CCWC.2018.8301729.
28. Sarkar A, Madhavidevi B, Nandi S, et al. Comparative machine learning approaches to identify the rice cultivars. 2024 2nd Int Conf Sustain Comput Smart Sys. 2024:945–50. https://doi.org/10.1109/ICSCSS60660.2024.10625290.
29. Karthik A. Deep learning-based plant disease classification using VGG-16: performance evaluation and comparative analysis. IEEE Int Conf Cont Comput Commun Materials. 2024;2024:1–7. https://doi.org/10.1109/ICCCCM61016.2024.11039860.
30. Barburiceanu S, Meza S, Orza B, et al. Convolutional neural networks for texture feature extraction. Application to leaf disease classification in precision agriculture. IEEE Access. 2021;9:160085–103. https://doi.org/10.1109/ACCESS.2021.3131002.
31. Ponmalar S, Devavaishnee M, Priya MI. Classification of rice varieties using VGG-19 and alexnet. 2023 IEEE 3rd Int Conf Smart Generation Comput Commu and Network. 2023:1–5. https://doi.org/10.1109/SMARTGENCON60755.2023.10442412.
32. Xu W, Fu YL, Zhu D. Resnet and its application to medical image processing: research progress and challenges. Comput Methods Programs Biomed. 2023;240:107660. https://doi.org/10.1016/j.cmpb.2023.107660.
33. Kamilaris A, Prenafeta-Boldú FX. Deep learning in agriculture: a survey. Comput Electron Agric. 2018;147:70–90. https://doi.org/10.1016/j.compag.2018.02.016.
34. Prasetyo SY, Lumentut JS. Enhancing rice variety identification through fine-tuning VGG-based deep learning models. 2024 IEEE 7th Int Conf Info Computational Sci. 2024:318–23. https://doi.org/10.1109/ICICoS62600.2024.10636881.
35. Mamun MAA, Riazul S, Karim I, et al. Evaluating the efficacy of hybrid deep learning models in rice variety classification. SSRN. 2024. https://ssrn.com/abstract=4749601.
36. Miftahushudur T, Sahin HM, Grieve B, et al. A survey of methods for addressing imbalance data problems in agriculture applications. Remote Sens. 2025;17(3):454. https://doi.org/10.3390/rs17030454.
37. Wu H. Maize leaf disease image classification based on resnet18 image classification model. 2023 Int Conf Comput Graphics Artif Intell Data Processing. 2024;13105:199–204. https://doi.org/10.1117/12.3026727.
38. Tran-Thi-Kim N, Pham-Viet T, Koo I, et al. Enhancing the classification accuracy of rice varieties by using convolutional neural networks. Int J Electrical Electron Eng Telecommun. 2023;12(2):150–60. https://doi.org/10.18178/ijeetc.12.2.150-160.
39. Khin O, Lee SK. Rice seed varieties classified using diffusion convolutional neural networks at various GPS locations. Sens Mater. 2025;37(6):2447–61. https://doi.org/10.18494/SAM5744.
40. Asif MJ, Khan H, Tehseen R, et al. Exploring convolutional neural networks for rice grain classification: an explainable AI approach. arXiv Preprint. arXiv:2025.05513. https://doi.org/10.48550/arXiv.2505.05513.
41. LaBreche HG, Nevins JR, Huang E. Integrating factor analysis and a transgenic mouse model to reveal a peripheral blood predictor of breast tumors. BMC Med Genomics. 2011;4(1):61. https://doi.org/10.1186/1755-8794-4-61.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.